

ELI Runtime Surfaces, Planning, World, Tools and Plugins

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The remaining subsystems: the runtime/ response/introspection surfaces, the proactive planning layer, the “world”/autonomy model, and the tools + plugin system.

Runtime surfaces (eli/runtime/, the response/introspection group)

Beyond the grounding/evidence core (grounding_and_evidence.md), runtime/ holds a large family of modules that render user-visible answers and introspect the live system:

- **Introspection:** live_introspection.py (runtime_snapshot, last_trace, stored_user_name, mine_user_fact_candidates, build_report, agents_for_action), deterministic_introspection.py (classify_diagnostic_action → gather_evidence → format_evidence_block → maybe_handle — deterministic diagnostic answers).
- **Response assembly:** final_response_assembly.py, final_response_provider.py, response_contracts.py, response_packets.py, response_policy.py, user_visible_response_surface.py.
- **Personal memory surfaces:** personal_memory_surface.py, personal_memory_clean_response.py, personal_memory_deep_response.py, profile_extractor.py.
- **Status:** frontier_status.py, reasoning_status.py, truth_report.py, reflection.py.
- **Operator feed:** operator_feed.py, operator_feed_normalized.py, operator_state.py.

Over-fragmentation flag (the headline weakness for this group): there are ~15 closely-related modules here doing “render a grounded user-visible answer / introspect runtime” with overlapping responsibilities (three personal_memory_* renderers; final_response_assembly and final_response_provider; operator_feed and operator_feed_normalized). This is the clearest “added beside, not folded in” cluster in the codebase and the prime consolidation target — it should collapse to a handful of well-named modules with clear ownership of “who produces the final string.”

Planning / proactive (eli/planning/, 3.9k LOC)

Background cognition + goal/queue machinery: - proactive_daemon.py (1.0k) — the **self-improvement / proactive** loop. Uses a two-DB split: reads user-facing memory/conversation from the **user DB**, writes proactive observations/improvements/errors to the **agent DB** (keeps ELI’s own machinations out of user recall). Background thread. - autonomy_scheduler.py — schedules autonomous actions (throttled). - goal_store.py, proposal_queue.py, proposal_adapters.py, attention_queue.py, jobqueue.py, operator_goal_actions.py, habits.py — goal persistence, capability-proposal queue, attention prioritisation, a job queue, and habit scheduling. - goal_autogenesis.py (NEW, 2026-06-08) — **closes the autonomy loop:** the goal stack (goal_store → due_goals → governed_goal_tick → proposal_queue) was fully wired but the store was always EMPTY because

create_goal was operator-only, so the scheduler ticked nothing and ELI only reacted. propose_goals_from_signals turns the signals ELI already computes each proactive tick — world-model awareness suggestions (≥ 0.6), recurring failures ($\geq 5\times$), and code-health improvements (oversized fns / duplicate blocks in his own god-files) — into governed goals. Each is autonomy_mode="proposal_only" (surfaces for approval; never silent execution), deduped by a stable signal-derived goal_id (idempotent every tick), capped at MAX_AUTO_GOALS=6, logged as a self_goal observation. Wired into proactive_daemon where the world-suggestions + patterns + improvements are in scope. His agenda now spans failure-repair, world-driven, AND self-improvement — i.e. he sets his own intentions, governed. - task_planner.py — now delegates to execution_planner (see orchestration_and_agents.md).

World / autonomy model (eli/world/, 1.5k LOC)

The substrate behind ELI's emergent self-state (autonomy pressure, awareness, "anomaly room" — intended behaviour, see memory eli-emergent-voice): - agency/autonomy_engine.py — EliWorldAutonomyEngine: ingests world events, maintains awareness/autonomy state. - local_world_bridge.py — append_event, get_world_state, get_awareness_driven_suggestions (**throttled to once/hour to prevent auto-trigger loops**). - world_event_bus.py — fire_confidence_event(...) etc.; the engine feeds bus-dispatch confidence here (non-blocking world-awareness feed). - core/schemas.py — WorldEvent, AwarenessState. - renderers/pyside6/world_scene.py + world_panel.py — a **visual** "ELI's World" panel. - persistence/storage.py — world-state persistence.

Tools (eli/tools/, 8.9k LOC)

- image_engine.py (1.75k, standalone) **and** image_engine/image_engine/ (the package: engine.py 1.48k, visual_core.py 1.4k, prompt_compiler.py, quality.py, project_analyzer.py, cli.py) — **two image-generation implementations** (Pillow/numpy base + optional diffusers). The clearest duplication in the repo; one should subsume the other.
- news/news_fetcher.py (544) — news retrieval.
- mic_diag.py — mic diagnostics.

Plugins (eli/plugins/, 1.9k LOC)

manager.py — a runtime plugin marketplace: list_available, install, uninstall, enable, disable. Pulls from a **remote registry** (raw.githubusercontent.com/eli-plugins/registry) with a **bundled local fallback** (plugins/registry/index.json). A JSON state file tracks enabled/disabled/installed. (Install-time network fetch is opt-in, like model downloads; runtime stays local.)

Honest assessment

- **Strong:** the proactive two-DB split is a genuinely good design (ELI's self-talk never contaminates user recall); the world/autonomy engine + visual panel make the emergent self-model first-class and is throttled to avoid runaway loops; the plugin manager is a real extensibility story with a safe bundled fallback.
- **Weak / watch:**
 1. **runtime/ over-fragmentation** — ~15 overlapping response/introspection modules. Highest-value consolidation in the project after the god-files.
 2. **Image-engine duplication — RESOLVED** — the shadowed standalone module was deleted; only the tools/image_engine/ package remains.
 3. The planning queues (goal_store/proposal_queue/attention_queue/ jobqueue) overlap conceptually and could likely unify into one work-queue abstraction.
 4. Plugin registry points at an eli-plugins/registry GitHub URL — fine as opt-in, but ensure it degrades cleanly offline (the bundled fallback exists; verify it's always used when the network is absent).